DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech **Branch: Common To All Branches** **Semester: I**

Subject Code & Name: BTES103 & Engineering Mechanics

Max Marks: 60 **Date:** 11/02/2025 **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

		(Level /CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**		**12**
1	A force system consists of: a) A single force acting on a body b) Two or more forces acting simultaneously on a body c) Only external forces d) Only internal forces	L1	1
2	Which of the following best represents the equilibrium condition for forces in two dimensions? a) $\Sigma Fx = 0, \Sigma Fy = 0, \Sigma Mz = 0$ b) $\Sigma Fx \neq 0, \Sigma Fy = 0, \Sigma Mz = 0$ c) $\Sigma Fx = 0, \Sigma Fy \neq 0, \Sigma Mz = 0$ d) $\Sigma Fx = 0, \Sigma Fy = 0, \Sigma Mx = 0$	L1	1
3	When two or more forces acting at a point are combined into a single force, it is called: a) Equilibrant b) Resultant force c) Couple d) Load factor	L1	1
4	The analytical conditions for equilibrium in two-dimensional force systems are: a) $\Sigma Fx = 0$ and $\Sigma Fy = 0$ b) $\Sigma Fx = 0, \Sigma Fy = 0$, and $\Sigma Mz = 0$ c) $\Sigma Fx \neq 0, \Sigma Fy = 0$ d) $\Sigma Mx = 0, \Sigma My = 0$	L1	1
5	In a force system in equilibrium, the sum of horizontal forces must be: a) Greater than vertical forces b) Equal to the sum of moments c) Equal to zero d) Less than vertical forces	L1	1
6	The force polygon method is used to: a) Find the equilibrium of parallel forces b) Determine the resultant force graphically c) Solve problems involving moments d) Find the center of gravity	L1	1

7 The equation used to determine if a truss is perfect: L1 1

a) m+4=2j b) m=2j−3

c) 2m=3+j d) m=j+r

8 When a body is just about to move, the frictional force acting on it is called: L1 1

a) Sliding friction b) Rolling friction

c) Limiting friction d) Dynamic friction

9 The slope of a velocity-time graph represents: L1 1

a) Displacement b) Acceleration

c) Velocity d) Force

10 When an object falls freely under gravity, its acceleration is: L1 1

a) Variable b) Zero

c) Constant and downward d) Constant and upward

11 According to D'Alembert's principle, the equation of motion can be written as: L1 1

a) F−ma=0 b) F+ma=0

c) F=ma d) F=m/v

12 The kinetic energy of a moving object depends on: L1 1

a) Velocity and mass b) Acceleration and mass

c) Force and displacement d) Work and power

Q. 2 Solve the following. **12**

A) State triangle law of forces and polygon law of forces. State and prove parallelogram law of forces. L2 6

B) What are various type of loadings? Distinguish clearly between uniformly distributed load, uniformly varying load and triangular load. L2 6

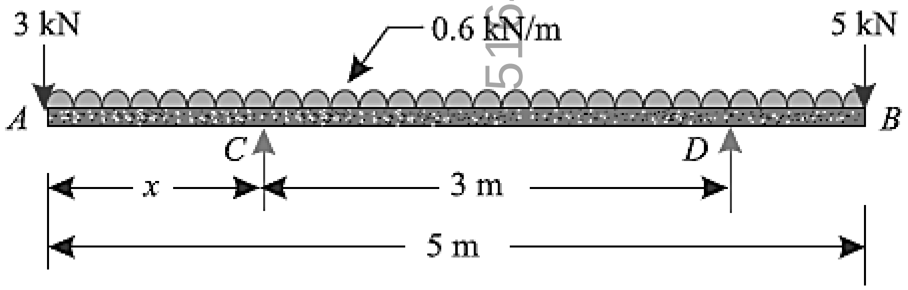
Q.3 Solve the following. **12**

A) An I-section has the following dimensions in mm units : L3 6

Bottom flange = 300 × 100, Top flange = 150 × 50, Web = 300 × 50

Determine mathematically the position of centroid of the section. Take bottom of the bottom flange as the axis of reference.

B) A beam AB 5 m long, supported on two intermediate supports 3 m apart, carries a uniformly distributed load of 0.6 kN/m. The beam also carries two concentrated loads of 3 kN at left hand end A, and 5 kN at the right hand end B. Determine the location of the two supports, so that both reactions are equal. L2 6

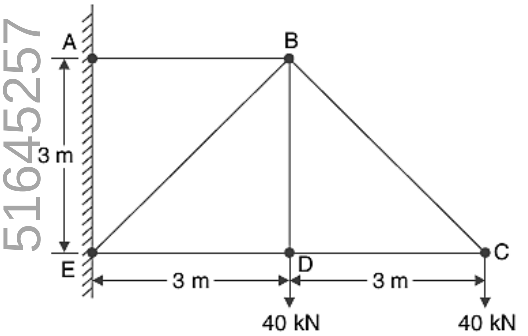


Q. 4 Solve Any Two of the following. **12**

A) Explain angle of friction, angle of repose and cone of friction with diagrams. L2 6

B) A body, resting on a rough horizontal plane, required a pull of 180 N inclined at 30° to the plane just to move it. It was found that a push of 220 N inclined at 30° to the plane just moved the body. Determine the weight of the body and the coefficient of friction. L3 6

C) Find the forces in all the members of the truss shown in figure. Tabulate the results. L2 6



Q.5 Solve Any Two of the following. **12**

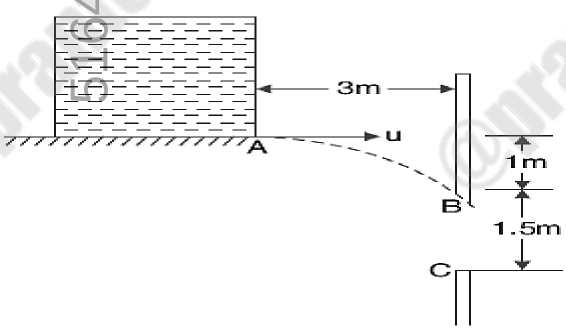
A) A stone dropped into a well is heard to strike the water in 4 seconds. Find the depth of the well, assuming the velocity of sound to be 335 m/sec. L2 6

B) Two ships leave a port at the same time. The first steams North-West at 32 kilometres per hour and the second 40° West to south at 24 kilometres per hour. L3 6

(a) What is the velocity of the second ship relative to the first in km per hour?

(b) After what time, they will be 160 km apart?

C) A pressure tank issues water at A with a horizontal velocity u as shown in figure. For what range of values of u, water will enter the opening BC. L2 6



Q. 6 Solve Any Two of the following. **12**

A) Describe a concept of mass moment of inertia. L2 6

B) A vehicle, of mass 500 kg, is moving with a velocity of 25 m/s. A force of 200 N acts on it for 2 minutes. Find the velocity of the vehicle: L2 6

(1) when the force acts in the direction of motion, and

(2) when the force acts in the opposite direction of the motion.

C) State and prove work energy principle. L3 6

***** End *****